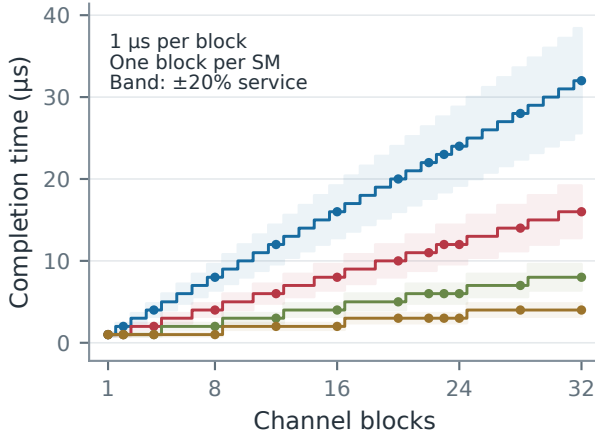


Channel work, shared resources, and finite slots

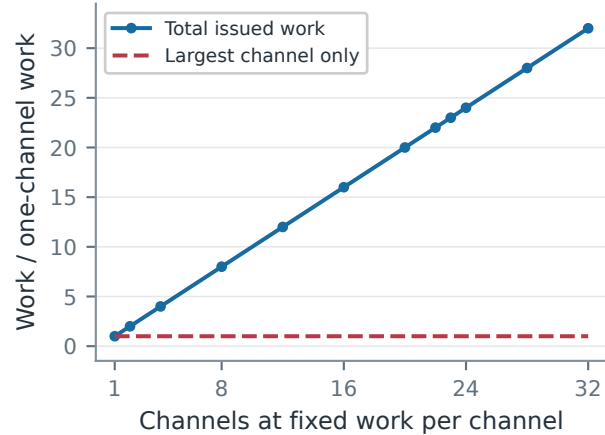
Controlled model checks with declared costs; no hardware calibration

— 1 SM — 2 SMs — 4 SMs — 8 SMs

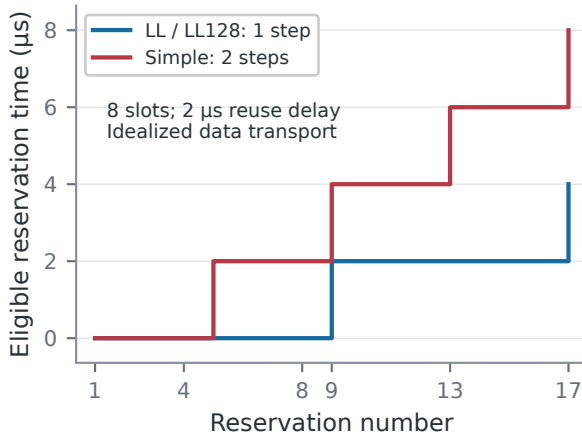
a Blocks retain finite GPU capacity



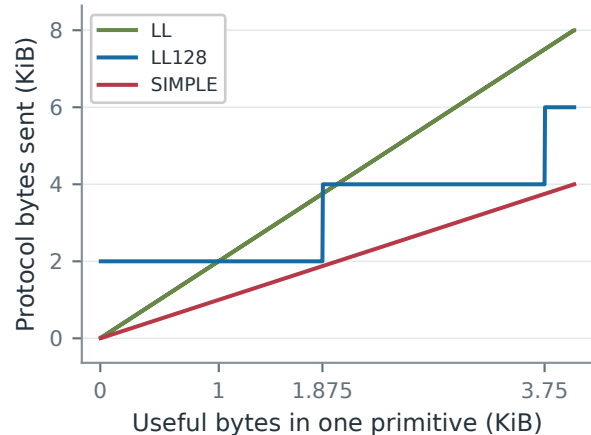
b Shared work grows with channels



c Reuse waits for the returned head



d LL128 sends full warp stripes



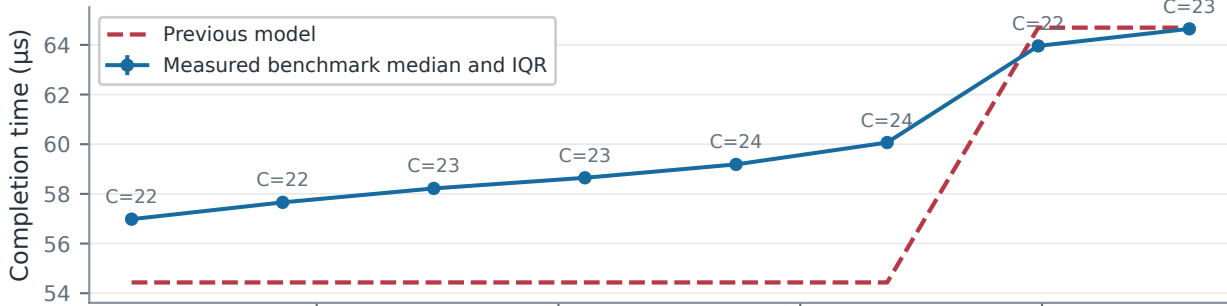
Exact residency and window oracles are separate from protocol-source byte checks.

LL128: each positive 1,920-byte warp stripe emits 2,048 bytes, including its partial tail.

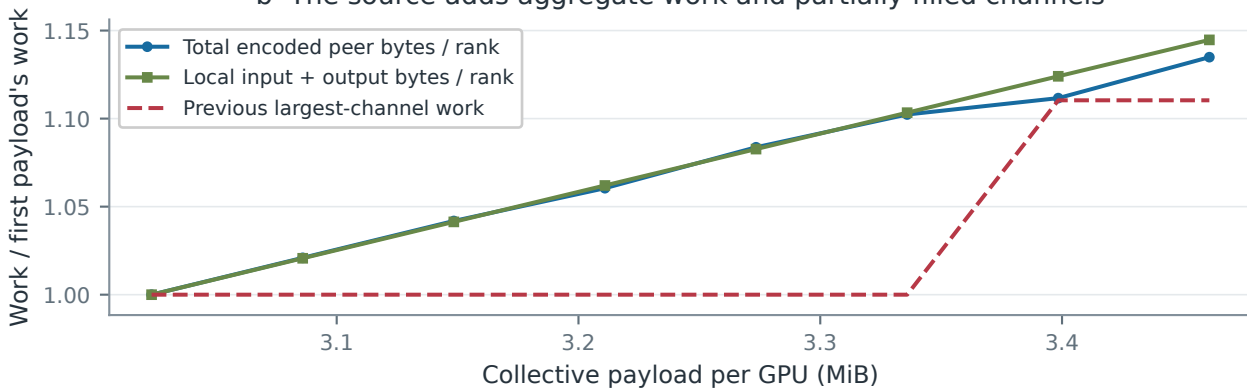
A100, four GPUs: the missing shared work

Retained LL128 measurements and new source-derived work counts

a Measured time rises while the previous prediction stays flat



b The source adds aggregate work and partially filled channels



These work counts repair an omitted mechanism; they do not yet identify its GPU service cost.

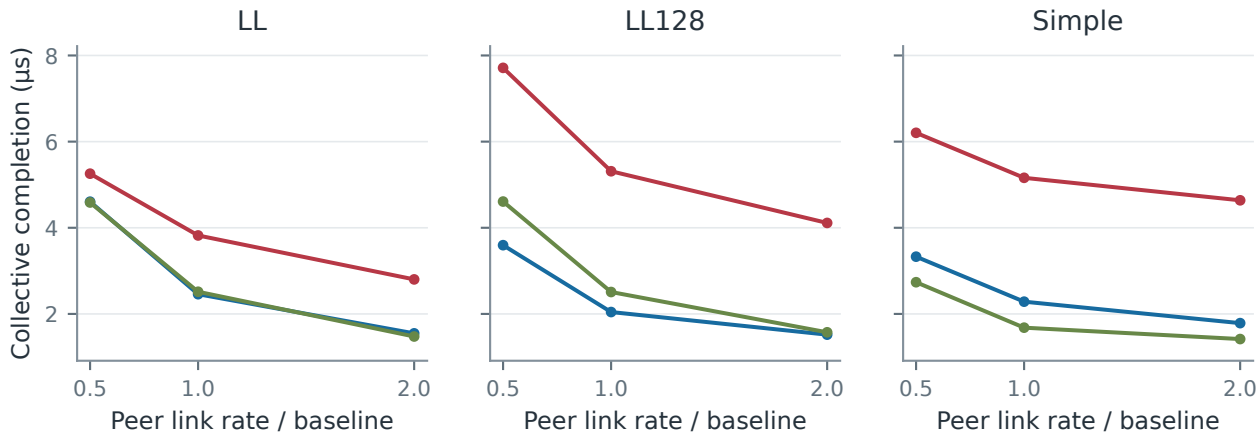
All points are retrospective inputs for the next model. No new fitted curve is shown here.

IQR: interquartile range across five independent processes. C: active channels.

GPU resources and links limit different parts of the same graph

Four ranks; 16 KiB per GPU; deterministic declared service inputs

—●— C=1, SMs=1 —●— C=4, SMs=1 —●— C=4, SMs=4

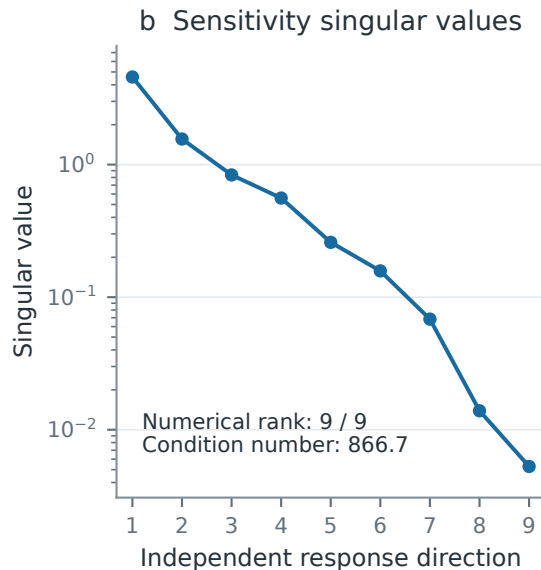
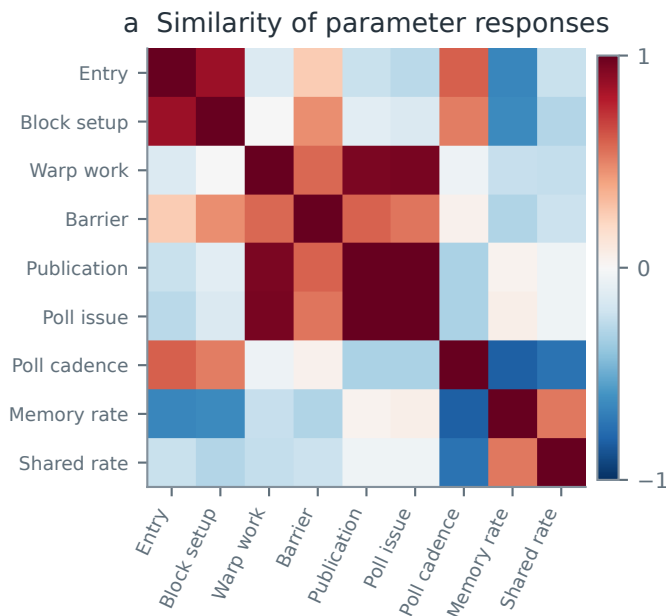


The physical calendar carries payload, protocol flags and separate progress stores.

This figure checks mechanisms. These are not calibrated A100 or GH200 predictions.

Which cost terms can these interventions distinguish?

Model sensitivity diagnostic, before any fit to new hardware observations



Rows vary protocol, two/four ranks, one/four channels and one/four available SMs.

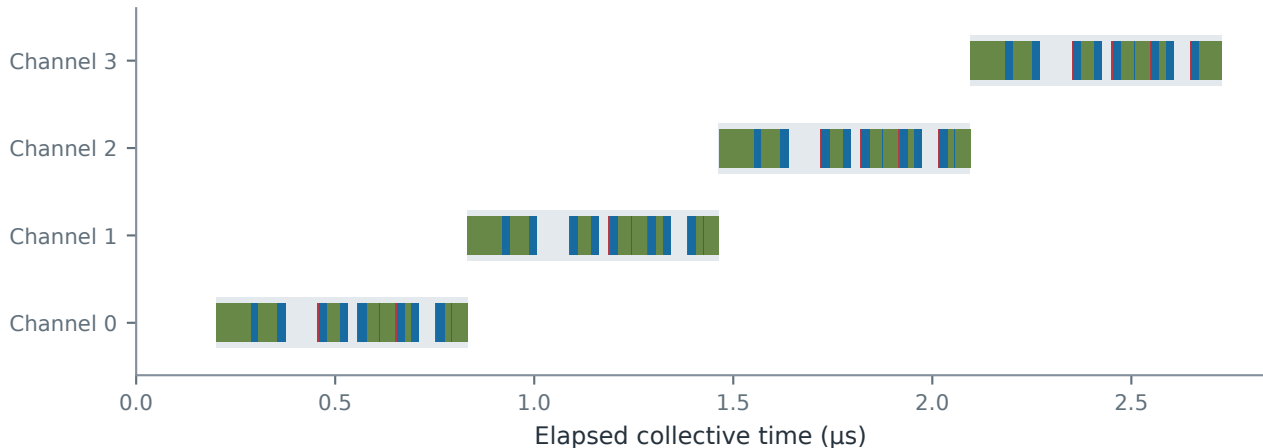
Full numerical rank does not establish physical identification. Similar responses amplify noise.

Instruction groups remain declared units; ready-peer and delayed-peer primitive probes are still needed.

One SM: waiting channels retain their block resources

GPU 0, four LL128 channels; all events use the physical transport calendar

Resident block Memory work Polling issue Other GPU work



Gray intervals include waiting. Colored spans show executed service, not an additive latency stack. GPU cycles are declared inputs. Network packets and returned capacity continue while a block waits.